

# Schnorr's Proof of Knowledge

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Protocols for Proof of knowledge

## 1 Schnorr's Dlog Proof of Knowledge

### 1.1 Protocol

- $g \rightarrow$  Group generator
- $p = \text{ord}(g)$

Prover ( $w, x = g^w$ )

- $r \xleftarrow{R} Z_p$
- $a = g^r$

$$z = r + c.w$$

$$\begin{array}{c} \longrightarrow \xrightarrow{a} \longrightarrow \\ \longleftarrow \xleftarrow{c} \longleftarrow \\ \longrightarrow \xrightarrow{z} \longrightarrow \end{array}$$

Verifier ( $x = g^w$ )

$$c \xleftarrow{R} Z_p$$

$$\text{Check } g^z \stackrel{?}{=} a.x^c$$

### 1.2 Properties

#### 1.2.1 Knowledge Extractability/ Soundness

The extractor runs the protocol twice while  $a$  remains the same in both runs.

Prover\*(malicious) ( $w, x = g^w$ )

$$\begin{array}{c} \longrightarrow \xrightarrow{a} \longrightarrow \\ \longleftarrow \xleftarrow{c^*} \longleftarrow \\ \longrightarrow \xrightarrow{z^*} \longrightarrow \end{array}$$

Extractor ( $x = g^w$ )

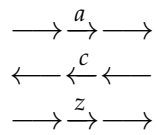
- $\frac{g^z}{g^{z^*}} = x^{c-c^*} =$
- $w = \frac{z-z^*}{c-c^*}$

### 1.2.2 Zero knowledge HVZK

HVZK because if  $V^*$  chooses  $c$  adaptively (not randomly), then Simulator is NOT Indistinguishable.

Simulator ( $w, x = g^w$ )

Verifier\*(mal) ( $x = g^w$ )



- $z \xleftarrow{R} \mathbb{Z}_p$
- $c \xleftarrow{R} \mathbb{Z}_p$
- $a = \frac{g^z}{x^c}$

### 1.2.3 Completeness

It is trivial.